**East West University**
Department of Computer Science & Engineering

# Lab Assignment 02

Self-Supervised Learning for Image Classification — BYOL & DINO

| | |
|---|---|
| **Course** | CSE 475: Machine Learning & Computer Vision |
| **Instructor** | Dr. Md Rifat Ahmmad Rashid, Associate Professor, EWU |
| **Total Marks** | 100 \| Kaggle Notebook (55) + Written Report (45) |
| **Deadline** | As announced on Google Classroom |
| **Restriction** | **ResNeXt architecture is NOT permitted** |

## Assignment Overview

### Purpose and Motivation

In **Assignment 01** you trained CNN and Transformer-based models—MobileNetV3, EfficientNet-B3, ResNeXt-50, ViT, Swin, and DeiT—on your group's assigned image dataset using fully supervised learning. While powerful, supervised learning has a fundamental bottleneck: acquiring large-scale, accurately labelled data is expensive, time-consuming, and impractical in many real-world domains such as medical imaging, satellite analysis, and industrial inspection.

**Assignment 02** addresses this limitation by introducing **Self-Supervised Learning (SSL)**—a paradigm that learns rich visual representations from *unlabelled* data by constructing auxiliary tasks from the data itself. You will implement and evaluate two state-of-the-art SSL frameworks:

- **BYOL** **Bootstrap Your Own Latent** (Grill et al., NeurIPS 2020) — a non-contrastive teacher–student approach that learns without negative pairs.
- **DINO** **Self-Distillation with No Labels** (Caron et al., ICCV 2021) — a self-distillation framework using a cross-entropy objective and a multi-crop augmentation strategy.

By completing this assignment you will:

1. Understand the theoretical foundations and architectural designs of BYOL and DINO.
2. Implement both frameworks on the Kaggle platform using the backbone from your best-performing Assignment 01 model.
3. Perform *linear probing* and *k-NN evaluation* to measure representation quality without fine-tuning.
4. Compare SSL performance against your Assignment 01 supervised baselines.
5. Produce a structured, professional lab report documenting every phase of the experiment.

## Connection to Assignment 01

**How Assignment 02 Builds on Assignment 01**

- **Same dataset.** You must use the *exact same assigned dataset* from Assignment 01. No new datasets are permitted.
- **Best model as backbone.** Identify the single model that achieved the highest test accuracy in Assignment 01 (CNN or Transformer). Extract its backbone (feature-extraction layers only, excluding the classification head) and use it as the encoder for both BYOL and DINO pre-training.
- **ResNeXt is excluded.** Even if ResNeXt-50 was your best-performing CNN in Assignment 01, you **must not** use it as the SSL backbone in this assignment. See Section 1.3 for full details and permitted alternatives.
- **Unlabelled pre-training protocol.** During SSL pre-training, you must **discard all labels**. Labels may only be reintroduced during linear probing and k-NN evaluation.
- **Supervised baselines.** The best CNN and Transformer accuracy figures from Assignment 01 serve as the performance reference for comparison in Task 4.
- **Same Kaggle dataset.** Re-use your group's uploaded Kaggle dataset from Assignment 01 as the data source.

## Architecture Restriction: ResNeXt is Not Permitted

**IMPORTANT NOTE**

**You may NOT use the ResNeXt-50 (or any ResNeXt variant) architecture as the backbone for SSL pre-training in this assignment.**

This restriction applies regardless of whether ResNeXt achieved the best accuracy in Assignment 01. The rationale is two-fold:

1. ResNeXt's grouped convolution design interacts poorly with the very small effective batch sizes that arise when BYOL/DINO use multi-crop augmentation, leading to unstable batch-normalisation statistics.
2. The assignment is designed to explore SSL on architectures you have not yet deeply analysed in Assignment 01—particularly EfficientNet and Vision Transformers.

**If ResNeXt was your Assignment 01 top model**, select the *second-best* model from your results table and use that backbone instead. Permitted backbone choices are listed in Section 1.4.

## Permitted Backbone Choices

The table below lists all permitted backbone encoders. Choose the one corresponding to your *best-performing non-ResNeXt model* from Assignment 01.

| Category | Backbone | Embedding Dim | Notes |
|---|---|---|---|
| CNN | MobileNetV3-Large | 960 | Lightweight; suitable for limited GPU RAM |
| CNN | EfficientNet-B3 | 1536 | Strong baseline; recommended for CNN track |
| ViT | ViT-S/16 | 384 | Recommended for DINO; native to SSL |
| ViT | Swin-T | 768 | Hierarchical ViT; good transfer features |
| ViT | DeiT-S | 384 | Distilled ViT; efficient and stable |

*ResNeXt-50 and all ResNeXt variants — PROHIBITED*

**Selection Rule**

1. Look up your Assignment 01 results table.
2. Find the model with the highest test accuracy that is **not** a ResNeXt variant.
3. Use that model's backbone (without its classification head) for both BYOL and DINO.
4. State your chosen backbone explicitly in the *Global Configuration* cell of your notebook and justify the choice in your written report.

# Theoretical Background

## Self-Supervised Learning — Core Concepts

Self-supervised learning creates *pseudo-supervision* automatically from the input data. For vision, the most successful family of methods uses **joint-embedding architectures**: two views of the same image are encoded and the model is trained so that the embeddings are similar irrespective of the augmentation applied.

### Data Augmentation Views

Both BYOL and DINO rely on augmentation to create diverse views of the same image. Each view is generated by randomly composing:

| Augmentation | Parameters |
|---|---|
| Random Resized Crop | Area $[0.08, 1.0]$; resized to $224 \times 224$ |
| Horizontal Flip | $p = 0.5$ |
| Colour Jitter | Brightness, contrast, saturation, hue |
| Grayscale | $p = 0.2$ |
| Gaussian Blur | $\sigma \in [0.1, 2.0]$ |
| Solarization (DINO) | Invert pixels above threshold, $p = 0.2$ |

## BYOL — Bootstrap Your Own Latent

### Architecture

BYOL uses two networks sharing the same backbone but with different update rules:

- **Online network** $f_\theta$: backbone → projector MLP → predictor MLP. Updated by gradient descent.
- **Target network** $f_\xi$: backbone → projector MLP (no predictor). Updated via **Exponential Moving Average (EMA)**:

$$\xi \;\leftarrow\; \tau\,\xi + (1-\tau)\,\theta, \qquad \tau \in [0.996,\ 1.0]$$

### Loss Function

Given two augmented views $v_1, v_2$ of the same image $x$:

$$\mathcal{L}_{\text{BYOL}} = -\tfrac{1}{2}\left[\frac{q_\theta(v_1)\cdot z_\xi(v_2)}{\|q_\theta(v_1)\|\,\|z_\xi(v_2)\|} + \frac{q_\theta(v_2)\cdot z_\xi(v_1)}{\|q_\theta(v_2)\|\,\|z_\xi(v_1)\|}\right]$$

The predictor and EMA target together prevent representational collapse without requiring negative pairs.

## DINO — Self-Distillation with No Labels

### Architecture

DINO uses a teacher–student structure where the teacher is an EMA copy of the student:

$$\theta_t \;\leftarrow\; \lambda\,\theta_t + (1-\lambda)\,\theta_s, \qquad \lambda \text{ follows a cosine schedule.}$$

### Multi-Crop Strategy

DINO generates multiple views per image:

- **2 global crops** at $224 \times 224$ (seen by teacher and student).
- **6–10 local crops** at $96 \times 96$ (seen only by student).

This asymmetric strategy forces the student to map local context to global semantics.

### Loss Function

$$\mathcal{L}_{\text{DINO}} = -\sum_{x \in \mathcal{V}_g} \sum_{\substack{x' \in \mathcal{V} \\ x' \neq x}} P_t(x) \log P_s(x')$$

where $P_t$ and $P_s$ are the teacher and student softmax distributions with temperatures $\tau_t < \tau_s$. A running-mean centering vector $c$ prevents collapse.

## Evaluation Protocols

| **Protocol** | Description |
|---|---|
| **Linear Probing** | Freeze the pre-trained backbone. Train a single `nn.Linear` layer on labelled data. Strong accuracy indicates linearly separable features. |
| **k-NN Classification** | Extract normalised embeddings for all training images. Classify test images by majority vote among the $k$ nearest neighbours. No additional training needed. |
| **Attention Visualisation** | (DINO only) Overlay self-attention maps from the last ViT block on input images to verify semantic grounding. |

## Tasks and Deliverables

### Task 1 — Dataset Preparation and EDA (10 marks)

**Task 1: Dataset Preparation and EDA**

1. **Reload** your Assignment 01 Kaggle dataset. Display the class distribution as a bar chart.
2. **Split** the data into:
   - 80% — unlabelled SSL pre-training pool (labels discarded).
   - 10% — labelled linear probe / k-NN training set.
   - 10% — held-out test set.
3. **Confirm label removal**: print the shape of the unlabelled pool and assert that no label tensor is passed to the SSL dataloader.
4. **Visualise** at least 16 augmented views of a single image, showing the effect of random crop, colour jitter, Gaussian blur, and solarization.
5. **Report**: number of images per split, per-channel mean and standard deviation, and class balance in the labelled split.

**Report Section: Dataset and Preprocessing (≈300–400 words)**

Include: class distribution plot, augmentation visualisation grid (16 views), split statistics table, and a brief justification of the 80/10/10 strategy.

### Task 2 — BYOL Pre-Training (25 marks)

## Task 2: BYOL Implementation and Pre-Training

**Backbone requirement (read carefully):**

**IMPORTANT NOTE**

Use the backbone of your **best-performing non-ResNeXt model** from Assignment 01 (see Section 1.4). **ResNeXt is prohibited.** Initialise the backbone weights from your saved Assignment 01 checkpoint, then replace the classification head with the BYOL projector and predictor.

1. Implement the BYOL online and target networks with the following heads:
   - **Projector MLP**: 3 layers, BatchNorm, hidden dim = 4096, output dim = 256.
   - **Predictor MLP**: 2 layers, BatchNorm, hidden dim = 4096, output dim = 256.
   - **EMA schedule**: $\tau$ from $0.996 \to 1.0$ (cosine).
2. Use AdamW optimiser (lr $= 3 \times 10^{-4}$, weight decay $= 1 \times 10^{-6}$).
3. Pre-train for a minimum of **100 epochs** on the unlabelled pool.
4. Plot and log the BYOL loss curve (epoch vs. loss).
5. Save backbone weights as `byol_backbone.pth`.

### Starter Code — BYOL Training Loop

```python
# -- BYOL TRAINING LOOP (pseudocode) ----------------------------------------
# NOTE: ResNeXt is NOT a valid choice for 'backbone'. Use your best A01 model
#       (MobileNetV3, EfficientNet-B3, ViT-S, Swin-T, or DeiT-S).

for epoch in range(NUM_EPOCHS):
    for images in unlabelled_loader:   # labels are NEVER loaded here
        v1 = augment_view(images)      # augmented view 1
        v2 = augment_view(images)      # augmented view 2

        # Online forward
        z1 = online_projector(online_backbone(v1))
        z2 = online_projector(online_backbone(v2))
        p1 = online_predictor(z1)
        p2 = online_predictor(z2)

        # Target forward (stop gradient)
        with torch.no_grad():
            zt1 = target_projector(target_backbone(v1))
            zt2 = target_projector(target_backbone(v2))

        # Symmetric negative cosine similarity loss
        loss = byol_loss(p1, zt2) + byol_loss(p2, zt1)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        # EMA update (online -> target)
        ema_update(online_params, target_params, tau=tau_schedule[epoch])
```

**Report Section: BYOL (≈400–500 words)**

Include: architecture diagram or pseudocode, hyperparameter table, loss curve plot, backbone choice justification, training time, and convergence analysis.

## Task 3 — DINO Pre-Training (25 marks)

**Task 3: DINO Implementation and Pre-Training**

**Backbone requirement (read carefully):**

**IMPORTANT NOTE**

Use the **same backbone** chosen in Task 2. If your best model was a CNN (e.g. EfficientNet-B3), you may use it for DINO by treating it as a plain encoder; a ViT-S/16 backbone is strongly recommended for DINO because the multi-head self-attention maps it produces enable the attention visualisation in Task 3, step 6. **ResNeXt remains prohibited.**

1. Implement DINO with:
   - **Projection head**: 3-layer MLP, $\ell_2$-normalised, output dim = 65 536.
   - **Multi-crop**: 2 global views at $224 \times 224$, 6 local views at $96 \times 96$.
   - **Teacher EMA**: $\lambda$ from $0.996 \to 1.0$ (cosine).
   - **Centering**: running-mean vector, momentum = 0.9.
   - **Temperatures**: $\tau_s = 0.1$; $\tau_t$ warmup from $0.04 \to 0.07$.
2. Use AdamW (lr $= 5 \times 10^{-4}$, weight decay cosine from $0.04 \to 0.4$).
3. Pre-train for a minimum of **100 epochs** on the unlabelled pool.
4. Plot and log the DINO loss curve.
5. Visualise DINO self-attention maps for at least **5 test images**, overlaying the last-block attention on the original image.
6. Save backbone weights as `dino_backbone.pth`.

### Starter Code — DINO Training Loop

```python
# -- DINO TRAINING LOOP (pseudocode) ----------------------------------------
# NOTE: ResNeXt is NOT permitted. Recommended: ViT-S/16 or EfficientNet-B3.

for epoch in range(NUM_EPOCHS):
    for images in unlabelled_loader:   # labels are NEVER loaded here
        # Multi-crop augmentation
        global_crops = [global_aug(images), global_aug(images)]
        local_crops  = [local_aug(images) for _ in range(N_LOCAL_CROPS)]

        # Teacher forward -- global views only, no gradient
        with torch.no_grad():
            teacher_out = [teacher(g) for g in global_crops]

        # Student forward -- all views
```

```
        student_out  = [student(v) for v in global_crops + local_crops]

        # DINO cross-entropy loss with centering
        loss = dino_loss(student_out, teacher_out,
                         center, tau_s=0.1, tau_t=tau_t_schedule[epoch])

        optimizer.zero_grad()
        loss.backward()
        clip_gradients(student.parameters(), max_norm=3.0)
        optimizer.step()

        # EMA teacher update + center update
        ema_update(student_params, teacher_params, lam=lam_schedule[epoch])
        center = update_center(center, teacher_out, momentum=0.9)
```

**Report Section: DINO (≈400–500 words)**

Include: architecture diagram, multi-crop strategy explanation, loss curve, 5 attention map figures (original + overlay), and semantic analysis of attended regions.

## Task 4 — Evaluation: Linear Probing and k-NN (25 marks)

**Task 4: Downstream Evaluation (backbone frozen throughout)**

### 4a. Linear Probing

1. Attach `nn.Linear(embed_dim, num_classes)` to the frozen backbone.
2. Train with SGD (lr = 0.01, momentum = 0.9) for 50 epochs.
3. Report: Top-1 accuracy, Top-5 accuracy (if classes $\geq 10$), confusion matrix, and per-class F1-score.
4. Repeat for **both** BYOL and DINO backbones.

### 4b. k-NN Classification

1. Extract $\ell_2$-normalised features for all labelled training and test images.
2. Evaluate for $k \in \{1, 5, 10, 20, 50, 200\}$ using cosine similarity.
3. Plot Accuracy vs. $k$ for BYOL and DINO on the same axes.

### 4c. Comparison Table

Populate the table below in your report (fill in your actual numbers):

| Method | Backbone | Epochs | Lin. Probe Top-1 | k-NN Acc. ($k$=20) |
|---|---|---|---|---|
| Supervised CNN (A01) | Best CNN (non-ResNeXt) | — | —% | — |
| Supervised ViT (A01) | Best ViT | — | —% | — |
| BYOL (ours) | Your chosen backbone | 100 | —% | —% |
| DINO (ours) | Your chosen backbone | 100 | —% | —% |

**Report Section: Evaluation and Results (≈500–600 words)**

Include: comparison table, 2 confusion matrices, k-NN accuracy vs. $k$ plot, per-class F1 bar chart, and a critical discussion of the gap between SSL and supervised accuracy on your specific dataset.

## Task 5 — Ablation, Discussion and Conclusion (15 marks)

**Task 5: Ablation Study (choose one option)**

1. **Augmentation ablation**: Remove one augmentation at a time (e.g. no Gaussian blur, no colour jitter) and measure the effect on linear probe accuracy for BYOL or DINO.
2. **EMA momentum ablation**: Train BYOL with $\tau \in \{0.99, 0.996, 0.999\}$ and compare linear probe accuracy.
3. **Label-fraction study**: Vary labelled fraction for linear probing as $\{1\%, 5\%, 10\%, 50\%\}$ and plot accuracy vs. label fraction for both methods.

**Report Section: Discussion and Conclusion (≈400–500 words)**

Cover: ablation results (plot or table), reflection on BYOL vs. DINO on your dataset, limitations (compute, data size, instability), and a concise conclusion summarising all findings. Include at least one paragraph on future directions (DINOv2, MAE, domain-specific fine-tuning).

# Kaggle Notebook Structure

**TWO Notebooks Required**

Students must submit **exactly two (2) separate Kaggle notebooks** for this assignment — one for **BYOL** and one for **DINO**. Both notebooks must be committed and made public on Kaggle before the deadline. Submitting only one notebook will result in **zero marks** for the missing model.

## Notebook 1 — BYOL

This notebook covers **Task 1 (shared EDA), Task 2 (BYOL pre-training), Task 4a–b (BYOL evaluation), and Task 5 (ablation)**.

### Required Section Structure

1. **[Markdown] Group Information** (mandatory first cell — see template below)
2. **[Code] Global Configuration** — all hyperparameters, backbone choice, random seed, paths
3. **[Code/MD] Setup and Imports**
4. **[Code/MD] Task 1 — Dataset EDA and Augmentation Visualisation**
5. **[Code/MD] Task 2 — BYOL: Model Definition** (online network, target network, projector, predictor)
6. **[Code/MD] Task 2 — BYOL: Pre-Training Loop** (unlabelled pool only; no labels used)
7. **[Code/MD] Task 2 — BYOL: Training Curve** (epoch vs. loss plot)
8. **[Code/MD] Task 4 — Linear Probing with BYOL Backbone**
9. **[Code/MD] Task 4 — k-NN Evaluation with BYOL Backbone**
10. **[Code/MD] Task 5 — Ablation Study**
11. **[MD] Conclusion**
12. **[MD] References**

### Notebook 1 Naming Convention

```
CSE 475 - Assignment 02 - BYOL - Group <ID>
```

## Notebook 2 — DINO

This notebook covers **Task 1 (shared EDA), Task 3 (DINO pre-training), Task 4a–b (DINO evaluation), and the comparison table (Task 4c)**.

### Required Section Structure

1. **[Markdown] Group Information** (mandatory first cell — same template)
2. **[Code] Global Configuration** — all hyperparameters, backbone choice, random seed, paths
3. **[Code/MD] Setup and Imports**
4. **[Code/MD] Task 1 — Dataset EDA and Augmentation Visualisation** (may reuse output from Notebook 1; must still appear here)
5. **[Code/MD] Task 3 — DINO: Model Definition** (student network, teacher network, projection head, multi-crop)
6. **[Code/MD] Task 3 — DINO: Pre-Training Loop** (unlabelled pool only; no labels used)
7. **[Code/MD] Task 3 — DINO: Training Curve** (epoch vs. loss plot)
8. **[Code/MD] Task 3 — DINO: Attention Map Visualisation** (at least 5 test images with self-attention overlay)
9. **[Code/MD] Task 4 — Linear Probing with DINO Backbone**
10. **[Code/MD] Task 4 — k-NN Evaluation with DINO Backbone**
11. **[Code/MD] Task 4 — Full Comparison Table** (BYOL vs. DINO vs. Assignment 01 supervised baselines)
12. **[MD] Conclusion**
13. **[MD] References**

### Notebook 2 Naming Convention

`CSE 475 - Assignment 02 - DINO - Group <ID>`

## Mandatory Group Information Cell (Both Notebooks)

The **very first cell** of **each** notebook must be a Markdown cell containing the following table. The `Notebook Type` field must reflect which notebook it is.

```
# CSE 475 - Assignment 02
## Group Information

| Field                        | Details                                        |
|-----------------------------|-----------------------------------------------|
| Group ID                     | Group XX                                       |
| Student 1 Name               | Full Name                                      |
| Student 1 ID                 | 20XX-X-XX-XXX                                  |
| Student 2 Name               | Full Name (if applicable)                      |
| Student 2 ID                 | 20XX-X-XX-XXX (if applicable)                  |
| Notebook Type                | BYOL Notebook  /  DINO Notebook                |
| Backbone Used                | e.g. EfficientNet-B3  (NOT ResNeXt)            |
| Assignment 01 Best Acc       | e.g. 87.3 % (EfficientNet-B3, 50 epochs)       |
| Dataset Name (Kaggle)        | /kaggle/input/<your-dataset-name>/             |
| Dataset Source               | e.g. Oxford-IIIT Pet Dataset                   |
```

```
| Dataset Source Link        | https://...                                |
| Submission Date            | DD Month 2026                              |
```

## Summary of Submission Requirements

| Item | Notebook Name | Contents | Marks |
|---|---|---|---|
| Notebook 1 | `...BYOL - Group XX` | Tasks 1, 2, 4 (BYOL), 5 | 30 |
| Notebook 2 | `...DINO - Group XX` | Tasks 1, 3, 4 (DINO + table) | 25 |
| PDF Report | `CSE475_A02_GroupXX_Report.pdf` | All tasks written up | 45 |
| **Grand Total** | | | **100** |

**IMPORTANT NOTE**

- Both notebooks must be **committed and public** on Kaggle before the deadline. Share both URLs via Google Classroom.
- Each notebook must run **end-to-end without errors**. Click *Run All* before committing.
- Notebooks named incorrectly will receive **zero marks** for the execution component.

## Written Report Structure

Submit a **PDF report (minimum 15 pages)** with the sections below. Use the headings exactly as listed.

| # | Section | Expected Content |
|---|---|---|
| 1 | Title Page | Assignment title, course, group info, backbone chosen, date |
| 2 | Table of Contents | Auto-generated |
| 3 | Introduction | Motivation for SSL; scope of BYOL and DINO; research questions |
| 4 | Dataset and Preprocessing | Dataset overview, EDA, split rationale, augmentation strategy |
| 5 | BYOL Implementation | Architecture, training config, loss curve, convergence analysis |
| 6 | DINO Implementation | Architecture, multi-crop strategy, loss curve, attention maps |
| 7 | Evaluation and Results | Linear probe, k-NN, comparison table, per-class analysis |
| 8 | Discussion and Ablation | Ablation results, reflection, limitations |
| 9 | Conclusion and Future Work | Summary of findings; directions for DINOv2, MAE etc. |
| 10 | References | IEEE format, minimum 8 citations |

**Formatting Requirements**

- Font: Times New Roman or equivalent serif, 11 pt.
- Margins: 2.5 cm on all sides.
- All figures must have captions and be referenced in the text.
- All tables must have titles.
- Equations must be numbered.
- File name: `CSE475_A02_Group<ID>_Report.pdf`

## Marks Distribution

| Task | Component | Notebook | Report |
|---|---|---|---|
| 1 | Dataset Preparation and EDA | 5 | 5 |
| 2 | BYOL Pre-Training | 15 | 10 |
| 3 | DINO Pre-Training | 15 | 10 |
| 4 | Evaluation (Linear Probe + k-NN) | 15 | 10 |
| 5 | Ablation, Discussion and Conclusion | 5 | 10 |
| | **Total** | **55** | **45** |
| | **Grand Total** | **100 Marks** | |

### Grading Rubric

| Range | Descriptor |
|---|---|
| **90–100** | All tasks complete; insightful analysis; ablation well-designed; report is professional with all figures, equations, and references. |
| **75–89** | All tasks complete; metrics correct; discussion present but lacking depth; minor formatting issues. |
| **60–74** | Most tasks complete; evaluation partially done; report missing figures or critical analysis. |
| **45–59** | At least one major task missing or incorrect; report is minimal. |
| **0–44** | Notebook does not run; major tasks absent; plagiarism detected. |

## Submission Guidelines

### What to Submit

1. **Kaggle notebook URL** — the committed, public notebook shared with the instructor via Google Classroom.
2. **PDF report** — submitted to Google Classroom before the deadline.

### Deadline Policy

**IMPORTANT NOTE**

- **Late penalty**: 10 marks deducted per 24-hour period after the deadline.
- Submissions more than 72 hours late will **not** be accepted.
- Extensions must be requested at least 48 hours in advance by email.

### Academic Integrity

**Academic Integrity Policy**

- You may consult open-source SSL libraries (e.g. `solo-learn`, `lightly`) as reference, but **all code must be written and adapted by your group** for your specific dataset and backbone.
- Copying code from other groups without attribution is academic dishonesty and results in **zero marks**.
- AI-generated report text used without substantive modification is prohibited.
- All references must be cited in IEEE format.

## Mandatory References

Your report must cite at least 8 references in IEEE format. The following are mandatory:

1. J.-B. Grill et al., "Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning," *NeurIPS 2020*.
2. M. Caron et al., "Emerging Properties in Self-Supervised Vision Transformers," *ICCV 2021*.
3. M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," *TMLR 2024*.
4. T. Chen, S. Kornblith, M. Norouzi, G. Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," *ICML 2020*.
5. K. He, X. Chen, S. Xie, Y. Li, P. Dollar, R. Girshick, "Masked Autoencoders Are Scalable Vision Learners," *CVPR 2022*.
6. A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," *ICLR 2021*.
7. *Citation for your Assignment 01 dataset* — full bibliographic details of the dataset you used.
8. *One additional recent SSL paper* from CVPR, ICCV, or NeurIPS 2022–2026 of your choice.

**Good Luck!**

This assignment brings you to the frontier of modern computer vision. Self-supervised learning is the technique behind the most powerful vision models deployed today at Meta AI, Google DeepMind, and beyond.
Take the time to *understand* each component, visualise intermediate outputs, and reflect critically on what your results reveal about your dataset.

**Dr. Md Rifat Ahmmad Rashid**
*Associate Professor, CSE — East West University*
*CSE 475: Machine Learning & Computer Vision*